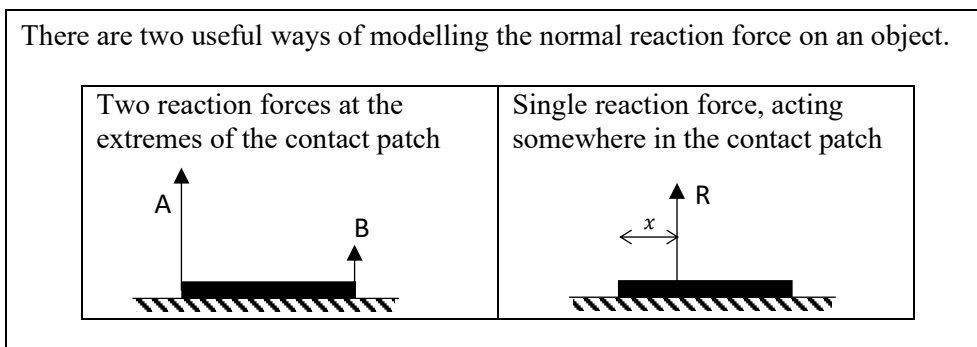
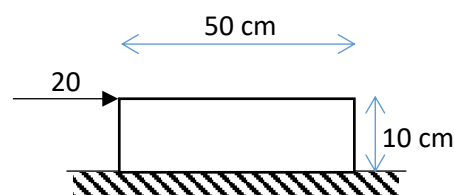


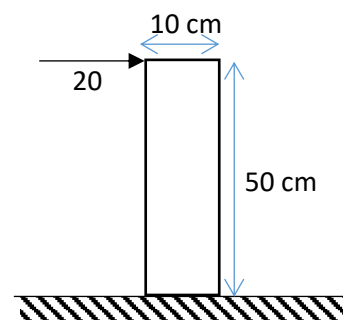


1. The diagram shows a uniform block of width 50 cm, height 10 cm, and mass 6 kg resting on rough horizontal ground. A force of 20 N acts as shown. The block does not slide.



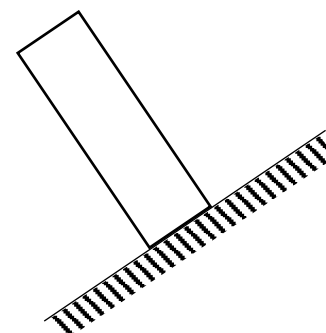
- Modelling the normal reaction using two forces A and B (one at each end of the block), draw a diagram showing all the forces acting on the block. Find A and B .
- How do you know from your answer to (a) that the block does not topple over?
- Draw a second diagram, this time modelling the reaction as a single force of R N, acting x cm from the left-hand end. Find R and x .
- How do you know from your answer to (c) that the block does not topple over?

2. The block is now placed on the rough plane in a different orientation, as shown.



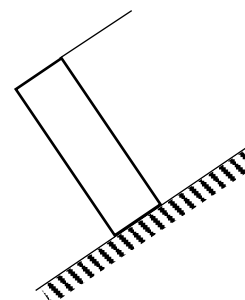
- Do the same as for Q1(a).
- How do you know from your answer to (a) that the block topples over?
- Do the same as for Q1(c).
- How do you know from your answer to (c) that the block topples over?

3. The block is now rotated through 90° and placed on a rough plane inclined at an angle of 30° . Model the reaction as a single force of R N, acting x cm from the lower end. Work geometrically to find R and x . Is the block stable?



4. In the situation of Q3, the angle of the slope is now gradually reduced until the block becomes stable. Find the angle at which this occurs.

5. In the situation of Q3, the block is now held in equilibrium by a string attached to the uppermost corner and angled parallel to the slope, as shown. Find the range of possible tensions in the string.

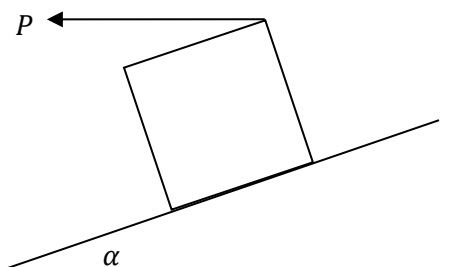


EQUILIBRIUM OF RIGID BODIES

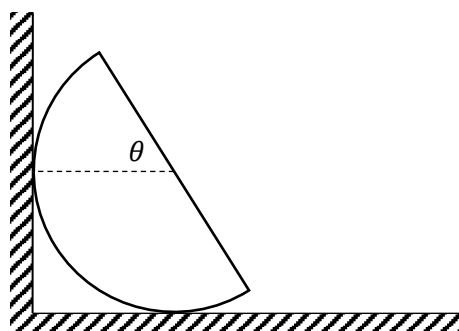
1. A uniform cube rests on a rough plane inclined at an angle α to the horizontal, where $\tan \alpha = \frac{1}{2}$. A gradually increasing horizontal force P is applied as shown in the diagram.

Show that equilibrium will be broken by sliding if $\mu < \frac{8}{11}$.

[ALTERNATIVE: find the condition for μ in terms of $\tan \alpha$, and verify that the above result holds when $\tan \alpha = \frac{1}{2}$.]



2. The diagram shows a solid hemisphere of radius a and weight W resting in limiting equilibrium against a smooth wall, and on rough horizontal ground. If the coefficient of friction between the hemisphere and the ground is μ then find an equation which relates θ and μ .



3. A sphere of radius a and weight W rests on horizontal ground. A thin uniform beam of weight $3\sqrt{3}W$ and length $2a$ is freely hinged to the ground at X , which is a distance $a\sqrt{3}$ from the point of contact of the sphere with the ground. The beam rests on the sphere, lying in the same vertical plane as the centre of the sphere. The coefficients of friction between the beam and the sphere and between the sphere and the ground are μ_1 and μ_2 respectively.
Given that the sphere is on the point of slipping at its contacts with both the ground and the beam, find the values of μ_1 and μ_2 .
4. A thin beam is fixed at a height $2a$ above a horizontal plane. A uniform straight rod ACB of length $9a$ and mass m is supported by the beam at C . Initially, the rod is held so that it is horizontal and perpendicular to the beam. The distance AC is $3a$, and the coefficient of friction between the beam and the rod is μ .

The rod is now released. Find the minimum value of μ for which B strikes the horizontal plane before slipping takes place at C .